

Proposed Formulation for HydroBoost - a Hydro-Hybrids Decision Support Tool

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1 Summary

This document introduces the mathematical formulation of the weekly profit maximization model for HydroBoost. It is expected to serve as the central optimization module for Subtask 2.3 of the project “Hydro + Storage: Accelerating Industry Deployment of Hydropower Hybrids”. This project focuses on developing a short-term investment decision tool/valuation tool for a hybrid co-located asset, which consists of a hydropower plant and a battery energy storage system (BESS). The research is being conducted by a team of experts from Idaho National Laboratory (INL), National Renewable Energy Laboratory (NREL) and Argonne National Laboratory (ANL). The objective is to provide an investment decision support tool that can demonstrate the technical and economic advantages of operating a hydroelectric plant in conjunction with a BESS.

2 Summary of model structure and assumptions

- A single hydroelectric plant with one or more hydro generators (turbine-generator sets)
- Neglecting net head effect of stored water over performance curve of generators
- Neglecting effect of hydraulic losses
- Hydro power generation function represented as non-convex piecewise linear function
- Rough zones represented by complementarity constraints
- Using concept of regulation signal to obtain quantitative estimation of net energy delivered by BESS in real time due to deployment of regulation up and down services [1, 2]
- Problem recast as Mixed Integer Linear Programming (MILP) model
- BESS model supported by [1, 2, 3, 4] and hydropower model by [5, 6]

Nomenclature

Sets

$\mathcal{T}$	Set of periods, where $\mathcal{T} = \{1, 2, \dots, N_H\}$
$\mathcal{U}$	Set of hydroelectric generators
$\mathcal{R}$	Set of non-dispatchable renewable power generators
$\mathcal{L}_j$	Set of linear segments in the performance curve of hydroelectric generator $j \in \mathcal{U}$
$\mathcal{Z}_j$	Subset of blocks or segments of hydroelectric generator j identified as rough zones of the unit, where $\mathcal{Z}_j \subseteq \mathcal{L}_j$

Parameters

N_H	Number of hours in short-term planning period
λ_t^E	Forecast market price of energy in hour t , at the bus or area where the hybrid asset is located [\$/MWh]
$\lambda_t^{RU}, \lambda_t^{RD}$	Forecast market price of regulation service (up and down) in hour t [\$/MW·h]
λ_t^{SR}	Forecast market price of spinning reserve SR in hour t [\$/MW·h]
ϵ	A very small number, such as $\epsilon = 0.0001$, incorporated into the objective function to avoid the simultaneous energy import and export
$\delta_t^{RU}, \delta_t^{RD}$	Fraction of regulation up (down) reserves sold in the day-ahead market, which are dispatched in real-time operation, where $\delta_t^{RU}, \delta_t^{RD} \in [0, 1]$
$p^{D,max}$	Maximum discharge capacity of BESS [MW]
$p^{C,max}$	Maximum charge capacity of BESS [MW]
RU^{max}	Maximum Regulation-Up that can be allocated to BESS [MW]
RD^{max}	Maximum Regulation-Down that can be allocated to BESS [MW]
SR^{max}	Maximum Spinning Reserve that can be allocated to BESS [MW]
η	Roundtrip efficiency of BESS in [p.u]
η^C	Charging efficiency of BESS in [p.u]
η^D	Discharging efficiency of BESS in [p.u]
SOC_t^{min}	Minimum state-of-charge of BESS in hour t [MWh]
SOC_t^{max}	Maximum state-of-charge of BESS in hour t [MWh]
SOC^{ini}	State-of-charge of BESS at beginning of period 1 [MWh]
I_t	Predicted water inflow to reservoir in hour t in [ft ³ /s]
SU_j, SD_j	Startup and shutdown cost of hydroelectric generator j [\$/h]
ρ_j^ℓ	Slope of block ℓ of performance curve of unit j [MW/(ft ³ /s)]

$\underline{U}_j$	Minimum water discharge of unit j [ft ³ /s]
$\overline{U}_j^\ell$	Maximum water discharge of block ℓ of unit j [ft ³ /s]
$P0_j$	Minimum power output of hydro generator j [MW]
$\underline{V}$	Minimum water storage level of reservoir [A-F]
$\overline{V}$	Maximum water storage levels of reservoir [A-F]
V^{ini}	Content of water in reservoir at beginning of period 1 [A-F]
φ	Fraction of initial volume of reservoir required at end of last period [p.u]
$\underline{\Theta}_t$	Total released water requirement in hour t [ft ³ /s]
W_j	Water needed to start up turbine-generator j [ft ³ /s]
$R_j^{\text{U,H}}, R_j^{\text{D,H}}$	Ramp up and ramp down of hydro generator j [MW/h]
$R_j^{\text{SU,H}}, R_j^{\text{SD,H}}$	Startup ramp and Shutdown ramp of hydro generator j [MW/h]
CF	Factor to convert water flow units in [ft ³ /s] into net volume units in [A-F]
C^{rz}	Penalization for operating within rough zone if honoring constraint is infeasible [$\$/(\text{ft}^3/\text{s})$]
M	A large enough scalar needed for representation of rough zone constraints
$\overline{T}^{(+)}$	Maximum capacity for transferring power to the grid for energy export [MW]
$\overline{T}^{(-)}$	Maximum capacity for transferring power from the grid for energy import [MW]

Variables

u_t^{B}	Binary variable driven to 1 when BESS is set to discharging mode, and 0 otherwise
$p_t^{\text{B,D}}$	Energy discharged from BESS and accounted for at point of delivery in period t [MWh]
$p_t^{\text{B,C}}$	Energy contributing to charge BESS in period t before accounting for losses [MWh]
p_t^{GB}	Energy from grid allocated to charge BESS (before considering any round-trip losses) [MWh]
e_t^{B}	Storage device state of charge in hour t [MWh]
$r_t^{\text{RU,D}}, r_t^{\text{RD,D}}$	Reserve for regulation up and regulation down, in discharging mode [MW]
$r_t^{\text{RU,C}}, r_t^{\text{RD,C}}$	Reserve for regulation up and regulation down, in charging mode [MW]
$r_t^{\text{RU}}, r_t^{\text{RD}}$	BESS reserve sold to market for regulation up and regulation down [MW]
$r_t^{\text{SR,D}}, r_t^{\text{SR,C}}$	Reserve for spinning reserve in discharging and charging modes [MW]
r_t^{SR}	BESS reserve sold to market for spinning reserve [MW]
u_{jt}^{H}	Binary variable which is equal to 1 if hydro generator j is on-line in hour t , and 0 otherwise

a_{jt}^H	Binary variable which is equal to 1 if hydro generator j is started at beginning of hour t , and 0 otherwise
z_{jt}^H	Binary variable which is equal to 1 if hydro generator j is shut down at beginning of hour t , and 0 otherwise
p_{jt}^H	Power output of hydroelectric generator j in hour t [MWh]
p_t^{HT}	Day-ahead total power setpoint of dispatchable hydroelectric plant in hour t [MWh]
p_t^{HB}	Hydroelectric power allocated to charge the BESS in hour t [MWh]
p_t^{HG}	Dispatchable hydroelectric power sold to the grid in the day-ahead market in hour t [MWh]
u_{jt}^ℓ	Water discharge of block ℓ of hydro generator j in hour t [ft ³ /s]
u_{jt}	Water discharge of hydro generator j in hour t [ft ³ /s]
w_{jt}^ℓ	Binary variable which is equal to 1 if water discharged by hydro generator j has exceeded block ℓ in hour t
s_t	Spillage of reservoir in hour t [ft ³ /s]
e_t^H	Water volume of reservoir in period t [ft ³]
$y_{jt}^{\ell(+)}, y_{jt}^{\ell(-)}$	Slack variables which take non-zero value if operation outside rough zone ℓ of unit j cannot be honored in hour t [ft ³ /s]
ψ_{jt}^ℓ	Auxiliary binary variable required for representation of rough zone constraint

3 Problem Formulation

3.1 Objective Function

The forecast profit of hybrid Hydro-BESS firm during week w is given by (1):

$$\Omega_w = \text{Max} \left\{ \sum_{t=1}^{N_H} \left(\Omega_t^E + \Omega_t^{AS} + \Omega_t^{RD} - \Phi_t^H \right) \right\} \quad (1)$$

Where:

- Ω_t^E Revenue from the sale of energy in the day-ahead energy market [\$]
- Ω_t^{AS} Revenue from sale of ancillary services in the day-ahead market clearing, including regulation-up, regulation-down and spinning reserve [\$]
- Ω_t^{RD} Adjustments to energy revenue from deployment of regulation up/down [\$]
- Φ_t^H Operational expenses of hydroelectric plant, formed by startup and shutdown costs plus penalties incurred if it is not possible to avoid operating within rough zone [\$]

The above terms are given as follows:

$$\Omega_t^E = \lambda_t^E \left[p_t^{\text{HG}} + p_t^{\text{B,D}} - (1 + \epsilon) p_t^{\text{GB}} \right] \quad (2)$$

$$\Omega_t^{AS} = \lambda_t^{\text{RU}} \cdot r_t^{\text{RU}} + \lambda_t^{\text{RD}} \cdot r_t^{\text{RD}} + \lambda_t^{\text{SR}} \cdot r_t^{\text{SR}} \quad (3)$$

$$\Omega_t^{RD} = \lambda_t^E \left(\delta_t^{\text{RU}} \cdot r_t^{\text{RU}} - \delta_t^{\text{RD}} \cdot r_t^{\text{RD}} \right) \quad (4)$$

$$\Phi_t^H = \sum_{j \in \mathcal{U}} \left(S U_j \cdot a_{jt}^H + S D_j \cdot z_{jt}^H + \sum_{\ell \in \mathcal{Z}_j} C^{\text{rz}} \left[y_{jt}^{\ell(+)} + y_{jt}^{\ell(-)} \right] \right) \quad (5)$$

In publications such as [3], a uniform price is set for regulation services. However, the proposed objective function aligns with the one used in the CHEERS model, as well as in [4] where distinct prices for regulation-up and regulation-down are considered. The term (3) varies, depending on market, as exemplified in [1, 4].

3.2 Constraints

3.2.1 Battery storage system

- (a) BESS energy balance equation, considering losses and regulation signal (deployment of up and down regulation):

$$e_t^B = e_{t-1}^B + \eta^C \cdot \left[p_t^{\text{B,C}} + \delta_t^{\text{RD}} \cdot r_t^{\text{RD,C}} - \delta_t^{\text{RU}} \cdot r_t^{\text{RU,C}} \right] - \frac{1}{\eta^D} \cdot \left[p_t^{\text{B,D}} + \delta_t^{\text{RU}} \cdot r_t^{\text{RU,D}} - \delta_t^{\text{RD}} \cdot r_t^{\text{RD,D}} \right] \quad t = 1, \dots, N_H \quad (6)$$

Notice that the balance equation includes the predicted deployment of regulation energy in real time, affected by losses.

(b) Limits on storage device state of charge, considering the provision of ancillary services:

$$e_{t-1}^B \geq \text{SOC}_t^{\min} + \frac{1}{\eta^D} (p_t^{B,D} + r_t^{\text{RU}} + r_t^{\text{SR}}) \quad t = 1, \dots, N_H \quad (8)$$

$$e_{t-1}^B \leq \text{SOC}_t^{\max} - \eta^C (p_t^{B,C} + r_t^{\text{RD}}) \quad t = 1, \dots, N_H \quad (9)$$

(c) Operational limits in discharging mode:

$$p_t^{B,D} + r_t^{\text{RU},D} + r_t^{\text{SR},D} \leq P^{D,\max} \cdot u_t^B \quad t = 1, \dots, N_H \quad (10)$$

$$p_t^{B,D} - r_t^{\text{RD},D} \geq 0 \quad t = 1, \dots, N_H \quad (11)$$

$$0 \leq r_t^{\text{RU},D} \leq \text{RU}^{\max} \quad t = 1, \dots, N_H \quad (12)$$

$$0 \leq r_t^{\text{RD},D} \leq \text{RD}^{\max} \quad t = 1, \dots, N_H \quad (13)$$

$$0 \leq r_t^{\text{SR},D} \leq \text{SR}^{\max} \quad t = 1, \dots, N_H \quad (14)$$

(d) Operational limits in charging mode:

$$p_t^{B,C} + r_t^{\text{RD},C} \leq P^{C,\max} \cdot (1 - u_t^B) \quad t = 1, \dots, N_H \quad (15)$$

$$p_t^{B,C} - r_t^{\text{RU},C} - r_t^{\text{SR},C} \geq 0 \quad t = 1, \dots, N_H \quad (16)$$

$$0 \leq r_t^{\text{RD},C} \leq \text{RD}^{\max} \quad t = 1, \dots, N_H \quad (17)$$

$$0 \leq r_t^{\text{RU},C} \leq \text{RU}^{\max} \quad t = 1, \dots, N_H \quad (18)$$

$$0 \leq r_t^{\text{SR},C} \leq \text{SR}^{\max} \quad t = 1, \dots, N_H \quad (19)$$

(e) Ancillary services offered to market:

(1) Reserve sold for Regulation Up:

$$r_t^{\text{RU}} = r_t^{\text{RU},D} + r_t^{\text{RU},C} \quad (20)$$

(2) Reserve for Regulation Down:

$$r_t^{\text{RD}} = r_t^{\text{RD},D} + r_t^{\text{RD},C} \quad (21)$$

(3) Reserve for Spinning Reserve:

$$r_t^{\text{SR}} = r_t^{\text{SR},D} + r_t^{\text{SR},C} \quad (22)$$

(f) Initial storage of BESS:

$$e_0^B = \text{SOC}^{\text{ini}} \quad (23)$$

3.2.2 Hydroelectric system (with reservoir)

(g) Commitment of hydroelectric generators:

$$u_{jt}^H - u_{j(t-1)}^H = a_{jt}^H - z_{jt}^H \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (24)$$

$$a_{jt}^H + z_{jt}^H \leq 1 \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (25)$$

$$u_{jt}^H, a_{jt}^H \in \{0, 1\} \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (26)$$

$$0 \leq z_{jt}^H \leq 1 \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (27)$$

(h) Generation of hydroelectric power:

$$p_{jt}^H = P0_j \cdot u_{jt}^H + \sum_{\ell \in \mathcal{L}_j} \rho_j^\ell \cdot u_{jt}^\ell \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (28)$$

$$p_t^{HT} = \sum_{j \in \mathcal{U}} p_{jt}^H \quad t = 1, \dots, N_H \quad (29)$$

(i) Water discharge of plant:

$$u_{jt}^1 \leq \bar{U}_j^1 \cdot u_{jt}^H \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (30)$$

$$u_{jt}^1 \geq \bar{U}_j^1 \cdot w_{jt}^1 \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (31)$$

$$u_{jt}^\ell \leq \bar{U}_j^\ell \cdot w_{jt}^{\ell-1} \quad \forall \ell \geq 2 \in \mathcal{L}_j, \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (32)$$

$$u_{jt}^\ell \geq \bar{U}_j^\ell \cdot w_{jt}^\ell \quad \forall \ell \geq 2 \in \mathcal{L}_j, \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (33)$$

$$u_{jt} = \sum_{\ell \in \mathcal{L}} u_{jt}^\ell + \underline{U}_j \cdot u_{jt}^H \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (34)$$

$$s_t - \sum_{j \in \mathcal{U}} W_j \cdot a_{jt}^H \geq 0 \quad t = 1, \dots, N_H \quad (35)$$

$$\sum_{j \in \mathcal{U}} u_{jt} + s_t - \underline{\Theta}_t \geq 0 \quad t = 1, \dots, N_H \quad (36)$$

(j) Water balance equation:

$$e_t^H = e_{t-1}^H + CF \left(I_t - \sum_{j \in \mathcal{U}} u_{jt} - s_t \right) \quad t = 1, \dots, N_H \quad (37)$$

(k) Ramping constraints:

$$p_{jt}^H - p_{j(t-1)}^H \leq R_j^{U,H} \cdot u_{j(t-1)}^H + R_j^{SU,H} \cdot a_{jt}^H \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (38)$$

$$-(p_{jt}^H - p_{j(t-1)}^H) \leq R_j^{D,H} \cdot u_{jt}^H + R_j^{SD,H} \cdot z_{jt}^H \quad \forall j \in \mathcal{U}, t = 1, \dots, N_H \quad (39)$$

(l) Water storage and spillage bounds:

$$\underline{V} \leq e_t^H \leq \bar{V} \quad t = 1, \dots, N_H \quad (40)$$

$$s_t \geq 0 \quad t = 1, \dots, N_H \quad (41)$$

(m) Initial and End-of-Period water volume constraints:

$$e_0^H = V^{\text{ini}} \quad (42)$$

$$e_{N_H}^H \geq \varphi \cdot V^{\text{ini}} \quad (43)$$

(n) Rough zone constraints:

$$\left(u_{jt}^\ell - y_{jt}^{\ell(-)}\right) \geq 0 \quad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \quad (44)$$

$$\left(\overline{U}_j^\ell - u_{jt}^\ell - y_{jt}^{\ell(+)}\right) \geq 0 \quad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \quad (45)$$

$$\left(u_{jt}^\ell - y_{jt}^{\ell(-)}\right) \leq M \cdot \psi_{jt}^\ell \quad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \quad (46)$$

$$\left(\overline{U}_j^\ell - u_{jt}^\ell - y_{jt}^{\ell(+)}\right) \leq M \cdot (1 - \psi_{jt}^\ell) \quad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \quad (47)$$

$$\psi_{jt}^\ell \in \{0, 1\} \quad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \quad (48)$$

$$y_{jt}^{\ell(+)}, y_{jt}^{\ell(-)} \geq 0 \quad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \quad (49)$$

$$y_{jt}^{\ell(+)}, y_{jt}^{\ell(-)} \leq \overline{U}_j^\ell \quad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \quad (50)$$

3.2.3 Coupling constraints

$$p_t^{\text{HT}} = p_t^{\text{HG}} + p_t^{\text{HB}} \quad t = 1, \dots, N_H \quad (54)$$

$$p_t^{\text{B,C}} = p_t^{\text{HB}} + p_t^{\text{GB}} \quad t = 1, \dots, N_H \quad (56)$$

3.2.4 Interconnection Limits

$$p_t^{\text{B,D}} + p_t^{\text{HG}} \leq \overline{T}^{(+)} \quad t = 1, \dots, N_H \quad (57)$$

$$p_t^{\text{GB}} \leq \overline{T}^{(-)} \quad t = 1, \dots, N_H \quad (58)$$

4 Regulation Signal

Regulation signal can be found at:

<https://www.pjm.com/markets-and-operations/ancillary-services>

Scroll down to **Regulation Self-Test Signals**. Choose the link and download file "Normalized Dynamic and Traditional Regulation Signals - May 2014", which is an .xls file. In that file, choose the Dynamic (RegD) worksheet.

5 Consideration of rough zones

Constraints (30)-(34) establish the dynamic of water release according to the operational region of the turbine-generator set. These operational regions are exemplified in Table 1 for a unit with a performance curve divided in 4 blocks, where for simplicity only the block number is indexed.

Table 1: Operating regions of unit with 4 blocks in performance curve

Block	Status of w^ℓ				Bounds on u^ℓ			
	w^1	w^2	w^3	w^4	u^1	u^2	u^3	u^4
1	0	0	0	0	$0 \leq u^1 \leq \bar{U}_1$	$u^2 = 0$	$u^3 = 0$	$u^4 = 0$
2	1	0	0	0	$u^1 = \bar{U}_1$	$0 \leq u^2 \leq \bar{U}_2$	$u^3 = 0$	$u^4 = 0$
3	1	1	0	0	$u^1 = \bar{U}_1$	$u^2 = \bar{U}_2$	$0 \leq u^3 \leq \bar{U}_3$	$u^4 = 0$
4	1	1	1	0	$u^1 = \bar{U}_1$	$u^2 = \bar{U}_2$	$u^3 = \bar{U}_3$	$0 \leq u^4 \leq \bar{U}_4$

Suppose that block 3 is identified as a rough zone for the turbine, so that generation within that range should be avoided. From Table 1, it is seen that this is possible by either keeping $u_3 = 0$ or $u_3 = \bar{U}_3$. This gives rise to the following complementarity constraint:

$$u_3 \cdot (\bar{U}_3 - u_3) = 0 \quad (59)$$

which can also be expressed as:

$$u_3 \geq 0 \perp (\bar{U}_3 - u_3) \geq 0 \quad (60)$$

If honoring the rough zone constraint is infeasible, the following slack variables are used, so that a solution by the MILP solver is always possible:

$$(u_3 - y_3^{(-)}) \geq 0 \perp (\bar{U}_3 - u_3 - y_3^{(+)}) \geq 0 \quad (61)$$

and $y_3^{(+)}$ and $y_3^{(-)}$ are penalized in the objective function, using a large scalar. After linearizing the above constraints with the "big M" method, these conditions are replaced by:

$$(u_3 - y_3^{(-)}) \geq 0 \quad (62)$$

$$(\bar{U}_3 - u_3 - y_3^{(+)}) \geq 0 \quad (63)$$

$$(u_3 - y_3^{(-)}) \leq \psi_3 \cdot M \quad (64)$$

$$(\bar{U}_3 - u_3 - y_3^{(+)}) \leq (1 - \psi_3) \cdot M \quad (65)$$

$$0 \leq y_3^{(-)} \leq \bar{U}_3 \quad (66)$$

$$0 \leq y_3^{(+)} \leq \bar{U}_3 \quad (67)$$

$$\psi_3 \in \{0, 1\} \quad (68)$$

where M is a large enough scalar.

References

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